

Quantum Communications and Networks(QU.50013)

Exercise Set 1

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Problem 1

The benefits of applying quantum technology come not from communication speeds, but rather the usefulness of sharing quantum states itself. **Quantum key distribution** (QKD) is one of them: to ensure computational security, we use the quantum mechanical principle that perfectly discriminating non-orthogonal states is impossible. Another is **quantum teleportation**, where unknown states can be sent far away, enabling distributed quantum computing systems. In both of these cases, qubits and ebits (entanglement) are the crucial quantum resources.

Problem 2

For a qubit in a pure state

$$|\psi\rangle = \frac{\sqrt{3}}{2} |0\rangle + \frac{1}{2} |1\rangle,$$

measurement probabilities can be determined using the Born rule:

$$P(0) = |\langle 0|\psi\rangle|^2 = \frac{3}{4}, \quad P(1) = |\langle 1|\psi\rangle|^2 = \frac{1}{4}.$$

Problem 3

An unknown quantum state cannot be simply read, copied, and retransmitted as classical information could. This is called the *no-cloning theorem*, and it stems from the fact that quantum measurement processes are destructive, i.e., measuring a system collapses its state into one of the measurement bases. Thus, quantum communication requires *quantum repeaters*, which help with retaining coherence in long-distance entanglements without amplifying signals.